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Smidt Heart Institute

Deep Learning-based Classification of Patients with Postural Orthostatic Tachycardia Syndrome using Wearable ECG and Accelerometer Data

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Introduction

Postural Orthostatic Tachycardia Syndrome (POTS) is a chronic autonomic disorder defined by orthostatic intolerance and ≥30 bpm HR increase without hypotension, with prevalence rising markedly during the COVID-19 pandemic. Conventional diagnostic methods like tilt-table or standing tests are limited by artificial conditions, patient discomfort, and inability to serve as general screening tools. This study evaluates wearable ECG and ACC data during daily activities, introducing a deep learning approach that detects posture-change points and classifies POTS versus controls..

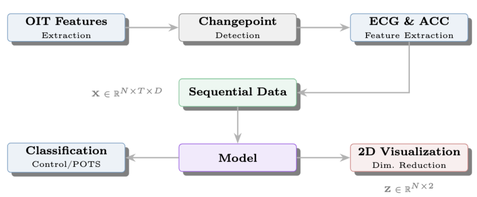

Figure 1: Wearable ECG and ACC features (X) are processed as sequential inputs (N subjects × T time steps × D features). The model performs end-to-end learning for classification of POTS vs. controls, and the learned representations (Z) are visualized in 2D after dimensionality reduction.

Methods

Study Population and Data Collection

The study analyzed 66 POTS and 20 control measurements under Cedars-Sinai IRB approval, with informed consent from all participants. Physiological data were continuously collected using the Faros 180 wearable monitor, recording 12-lead ECG at 1 kHz and triaxial ACC at 25 Hz for at least 24 hours. Participants performed their usual daily activities without restrictions, ensuring that the data accurately reflected natural physiological routines.

Data Preprocessing and Feature Extraction

Data preprocessing included noise reduction of ECG (0.5–150 Hz band-pass) and accelerometer signals, with quality assessed using SNR, template matching, and HRV metrics. Accelerometer data were transformed via Heuristic Orientation-Invariant Transformation (OIT), and posture-change events were detected using the PELT algorithm. Around each changepoint, ECG features and ACC features were extracted.

Feature Category	Key Characteristics
ECG-based Features	
Statistical	Amplitude statistics, autocorrelation, trend analysis, Hjorth parameters
Spectral	Power spectrum, frequency entropy, wavelet decomposition
Morphological	Zero crossings, peak-trough metrics, waveform shape
Physiological	Heart rate change, SKNA burst
ACC-based Features	
Movement Intensity	Magnitude, jerk, SMA, RMS, energy metrics
Postural	Stability measures, dynamic acceleration, OIT statistics
Spatial	Rotation vector, trajectory curvature, movement patterns
Temporal	Decay time, spectral entropy, windowed analysis

Table 1: Summary of Feature Categories

Model Development and Training

Two baseline deep learning models were implemented: (1) a Transformer with ALiBi attention (6 layers, hidden size 128, 8 heads, dropout 0.3) and (2) a CNN-BiLSTM-Attention hybrid combining convolution, bidirectional LSTM, and self-attention layers. Training used Adam optimizer (lr = 1e-4, weight decay = 1e-4) with focal loss to address class imbalance. Data augmentation (jittering, scaling, permutation), gradient clipping, and learning rate scheduling were applied to enhance generalization and stability. Model performance was evaluated with 10-fold stratified cross-validation and a separate holdout set. Metrics included balanced accuracy, sensitivity, specificity, precision, recall, F1-score, and AUC. Confusion matrices and learned feature visualizations were also generated.

Measure	Formula	Description
SNR	$10\log_{10}(P_{\text{signal}}/P_{\text{noise}})$	Signal-to-noise ratio (power ratio in dB)
TM	$\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum(x_i-\bar{x})^2\sum(y_i-\bar{y})^2}}$	Morphological similarity (template x_i, beat y_i)
RMSSD	$\sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(RR_{i+1}-RR_i)^2}$	RR variability (successive RR intervals)
SDNN	$\sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(RR_i-\overline{RR})^2}$	RR variability (standard deviation)
PE	$-\sum_i p_i \ln(p_i)$	Complexity entropy (p_i: ordinal probability)

Table 2: ECG Signal Quality and Variability Metrics

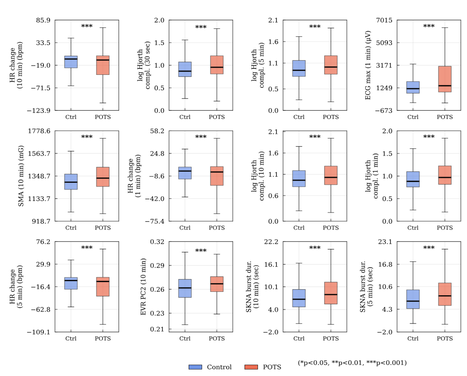

Figure 2: Comparison of physiological variables between Control and POTS groups

Results

Figure 2 shows the comparison of key physiological features between POTS patients and controls. Significant group differences were observed across multiple variables, including heart rate change, SKNA burst duration, SMA, ECG amplitude, and Hjorth complexity measures. These results highlight distinct physiological patterns around posture-change events, supporting the selection of discriminative features for model development.

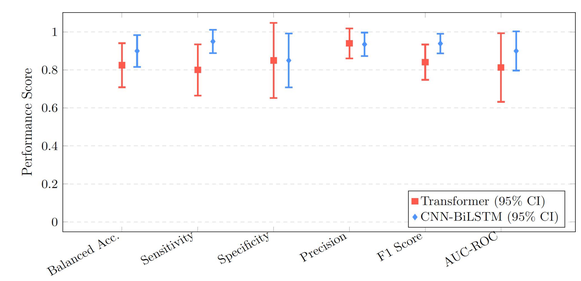

Figure 3: 10-Fold Cross Validation Results with 95% Confidence Intervals: Model performance comparison with mean values and confidence intervals.

Figure 3 summarizes the 10-fold cross-validation results. The CNN-BiLSTM-Attention model achieved superior balanced accuracy (0.900 ± 0.084), sensitivity (0.950 ± 0.062), F1 score (0.939 ± 0.052), and AUC (0.900 ± 0.103) compared to the Transformer, highlighting its stronger performance in classifying POTS patients versus controls.

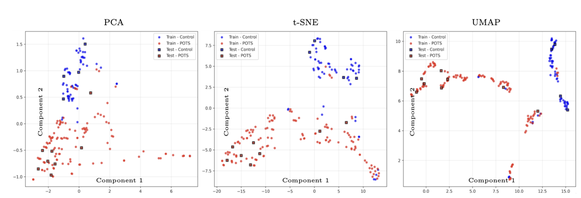

Figure 4: Feature Space Visualization - CNN-BiLSTM-Attention Model.

Figure 4 illustrates the feature space visualization of the CNN-BiLSTM-Attention model. Using PCA, t-SNE, and UMAP, the learned latent representations show clear separation between POTS patients and controls. Test samples from POTS patients aligned closely with the training distribution, demonstrating the model's robust representation learning and generalization ability.

Discussion

- The findings suggest potential clinical implications for enhancing the diagnostic approach to POTS, with wearable-based methodology enabling continuous and unobtrusive monitoring of ECG and ACC features in real-world settings.
- Deep learning models trained on wearable sensor data demonstrated promising performance in differentiating between POTS patients and healthy controls, but additional validation with expanded cohorts and varied clinical settings is required.
- The dataset may not fully represent the diversity of clinical conditions and populations, highlighting the need for larger and more representative samples to improve generalizability.
- Future research should integrate additional physiological parameters or environmental factors to enhance model robustness and clarify whether classifiers detect disease-specific signals or patterns influenced by medication use.
- ECG and ACC-derived features analyzed in this study may hold diagnostic promise beyond POTS, with prior studies showing utility in autonomic dysfunction, arrhythmias, diabetes, Parkinson's disease, and fall risk assessment.
- These findings indicate the potential for broadly applicable diagnostic technologies targeting diverse neurological and autonomic conditions.

Conclusion

- This study investigated the feasibility of using deep learning with wearable-derived ECG and ACC data to differentiate POTS patients from healthy controls.
- The wearable-based approach captured subtle, clinically meaningful physiological variations and symptom patterns in real-world daily life.
- It demonstrated potential to address limitations of conventional diagnostic methods such as the tilt-table test.
- Two deep learning models—Transformer and hybrid CNN-BiLSTM—showed promising performance for classification.
- The integration of wearable signal analysis and deep learning suggests promise for patient-centered diagnostics.
- Overall, this framework supports improved clinical management for individuals with POTS.

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